

Team Application Exercises (tAPP-6) – solutions

Problem 1: For each of the following Java code fragments, indicate exactly what will happen when it is compiled and executed. If it produces output, show the exact output; if it runs but produces an error, specify the error precisely; if it will not compile, describe what the problem is.

(ii) **public class A {**
 public int i = 3;

public A(int val) {
 int i = val;
 }
}

A a = **new** A(5);
System.out.println(a.i);

Answer: 3

(Because inside the constructor,
i refers to the local variable, not the
field)

(iii) **int i = 5.0;**
 int j = 10.0;
System.out.println(i / j);

Answer: ClassCastException

(Because you can't assign a **double**
value to an **int** variable)

(iv) **try {**
 throw new Exception("Hello!");
 System.out.println("World!");
} catch (Exception ex) {
 System.out.println(ex.getMessage());
}

Answer: error because the first
println statement is unreachable

(v) *// File A.java*
 public class A {
 public A(int value) {
 System.out.println("Value is " + value);
 }
 }

// File B.java
 public class B extends A {
 public static void main (String[] args) {
 A a = **new** A(10);
 }
 }

File B.java will not compile: Class B has no
constructor defined, so it will attempt to call
the nonexistent no-args constructor of A.

Problem 2: You are to design a class to model the employees of a company. Every employee has an employee number, a family name, and a given name. In addition, each employee has a basic hourly rate, a number of contracted hours per month, and a number of hours actually worked (in the current month). Workers are paid 1.5 times their normal rate for any hours above their contracted number.

Write a class definition for **Employee**, incorporating all of the attributes mentioned above. Be sure to use appropriate data types and access modifiers. You should also define a public constructor for **Employee**, which should initialise all fields – the initial value for “hours actually worked” should be zero. You do not need to define any instance methods.

```
public class Employee {  
  
    private String familyName;  
    private String givenName;  
    private int employeeNum;  
  
    private double hourlyRate;  
    private int contractedHours;  
    private int actualHours;  
  
    public Employee(String familyName, String givenName, int employeeNum,  
                    double hourlyRate, int contractedHours)  
        throws IllegalArgumentException  
    {  
        this.familyName = familyName;  
        this.givenName = givenName;  
        this.employeeNum = employeeNum;  
  
        this.contractedHours = 0;  
  
        if (hourlyRate <= 0) {  
            throw new IllegalArgumentException("Invalid hourly rate: " + hourlyRate);  
        }  
        this.hourlyRate = hourlyRate;  
  
        if (contractedHours < 0) {  
            throw new IllegalArgumentException("Invalid contracted hours: " + contractedHours);  
        }  
        this.contractedHours = contractedHours;  
    }  
}
```

Problem 3: Using the same **Employee** definition as above, write a method **computePay()** that computes and returns the employee’s pay for the current month based on the current values of the appropriate fields.

```
public double computePay() {  
    double extraHours = Math.max(actualHours - contractedHours, 0);  
  
    double regularHours = Math.min(contractedHours, actualHours);  
  
    return (regularHours * hourlyRate) + (1.5 * extraHours * hourlyRate);  
}
```